

SECURE BOOT AND DATA ENCRYPTION: AN ECC- INTEGRATED AUTHENTICATION PROTOCOL FOR EMBEDDED DEVICES

VELLORE INSTITUTE OF TECHNOLOGY — CHENNAI CAMPUS

COURSE -

BCSE305L – EMBEDDED SYSTEMS

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PROBLEM STATEMENT

WHY EMBEDDED SECURITY IS BROKEN TODAY

EMBEDDED SYSTEMS AND IOT DEVICES ARE DEPLOYED IN THE MOST SECURITY-CRITICAL ENVIRONMENTS — MEDICAL DEVICES, INDUSTRIAL AUTOMATION, SMART HOMES, AND CONNECTED VEHICLES — YET THEY REMAIN ALARMINGLY

❌ CONVENTIONAL RSA CRYPTOGRAPHY IS COMPUTATIONALLY INFEASIBLE ON MICROCONTROLLERS WITH KILOBYTES OF RAM AND MILLIWATTS OF POWER BUDGET.

PROPOSED SOLUTION

IMPLEMENT AN ECC-BASED AUTHENTICATION AND ENCRYPTION PROTOCOL OPTIMIZED FOR EMBEDDED DEVICES — DELIVERING STRONG SECURITY WITH MINIMAL RESOURCE OVERHEAD.

LIMITED COMPUTE

CANNOT EFFICIENTLY RUN RSA ($O(K^3)$ COMPLEXITY) ON MICROCONTROLLERS

MEMORY CONSTRAINTS

KILOBYTES OF RAM INSUFFICIENT FOR LARGE RSA KEY STORAGE

POWER LIMITATIONS

BATTERY-POWERED DEVICES DEMAND ENERGY-EFFICIENT CRYPTOGRAPHY

WIRELESS VULNERABILITIES

OPEN-AIR CHANNELS EXPOSED TO EAVESDROPPING AND MITM ATTACKS

PHYSICAL TAMPERING

SIDE-CHANNEL ATTACKS AND UNAUTHORIZED FIRMWARE MODIFICATION

NO STANDARD AUTH

LACK OF ROBUST DEVICE-TO-DEVICE AUTHENTICATION PROTOCOLS

OBJECTIVES & GAP ANALYSIS

WHAT WE BUILD — AND WHY EXISTING APPROACHES FALL SHORT

PROJECT OBJECTIVES

- 01
- LIGHTWEIGHT KEY EXCHANGE

DESIGN ECDH PROTOCOL OPTIMIZED FOR EMBEDDED PLATFORMS
- 02
- AUTHENTICATED ENCRYPTION

INTEGRATE AES-256-GCM WITH BUILT-IN INTEGRITY VERIFICATION
- 03
- DIGITAL SIGNATURES

CHALLENGE-RESPONSE AUTHENTICATION VIA ECDSA
- 04
- SECURE BOOT

FIRMWARE INTEGRITY VERIFICATION BEFORE EXECUTION
- 05
- HW BENCHMARKING

PERFORMANCE METRICS ON ESP32 AND RASPBERRY PI

EXISTING SYSTEM LIMITATIONS

APPROACH	LIMITATION
PASSWORD-BASED AUTH	VULNERABLE TO BRUTE-FORCE; UNSUITABLE FOR DEVICE-TO-DEVICE COMMUNICATION
RSA CRYPTOGRAPHY	$O(K^3)$ COMPLEXITY; 2048–4096-BIT KEYS; EXCESSIVE MEMORY AND POWER DEMAND
TLS / DTLS	HEAVYWEIGHT; IMPLEMENTATIONS EXCEED AVAILABLE RAM ON MICROCONTROLLERS
BLE SECURITY	LIMITED RANGE; RELIES ON PRE-SHARED KEYS OR OUT-OF-BAND PAIRING

PROPOSED SYSTEM

ECC-BASED SECURE COMMUNICATION PIPELINE

A COMPLETE END-TO-END CRYPTOGRAPHIC STACK COMBINING SIX TIGHTLY INTEGRATED COMPONENTS:



ECC P-256

256-bit curves equivalent to 3072-bit RSA — at **1/12th the key size** and $O(\log n)$ complexity



ECDH KEY EXCHANGE

SECURE SHARED SECRET DERIVATION OVER UNTRUSTED WIRELESS CHANNELS



HKDF-SHA256

CRYPTOGRAPHICALLY STRONG SESSION KEY DERIVATION FROM SHARED SECRET



AES-256-GCM

AUTHENTICATED ENCRYPTION WITH BUILT-IN INTEGRITY TAG VERIFICATION



ECDSA SIGNATURES

MUTUAL DEVICE AUTHENTICATION VIA CHALLENGE-RESPONSE PROTOCOL



SECURE BOOT

FIRMWARE INTEGRITY VERIFIED CRYPTOGRAPHICALLY BEFORE EXECUTION

⚡ 100–1000× faster

ECC $O(\log N)$ VS. RSA $O(K^3)$ — ORDERS OF MAGNITUDE MORE EFFICIENT



90% SMALLER KEYS

256-bit ECC delivers proven security equivalent to 3072-bit RSA keys

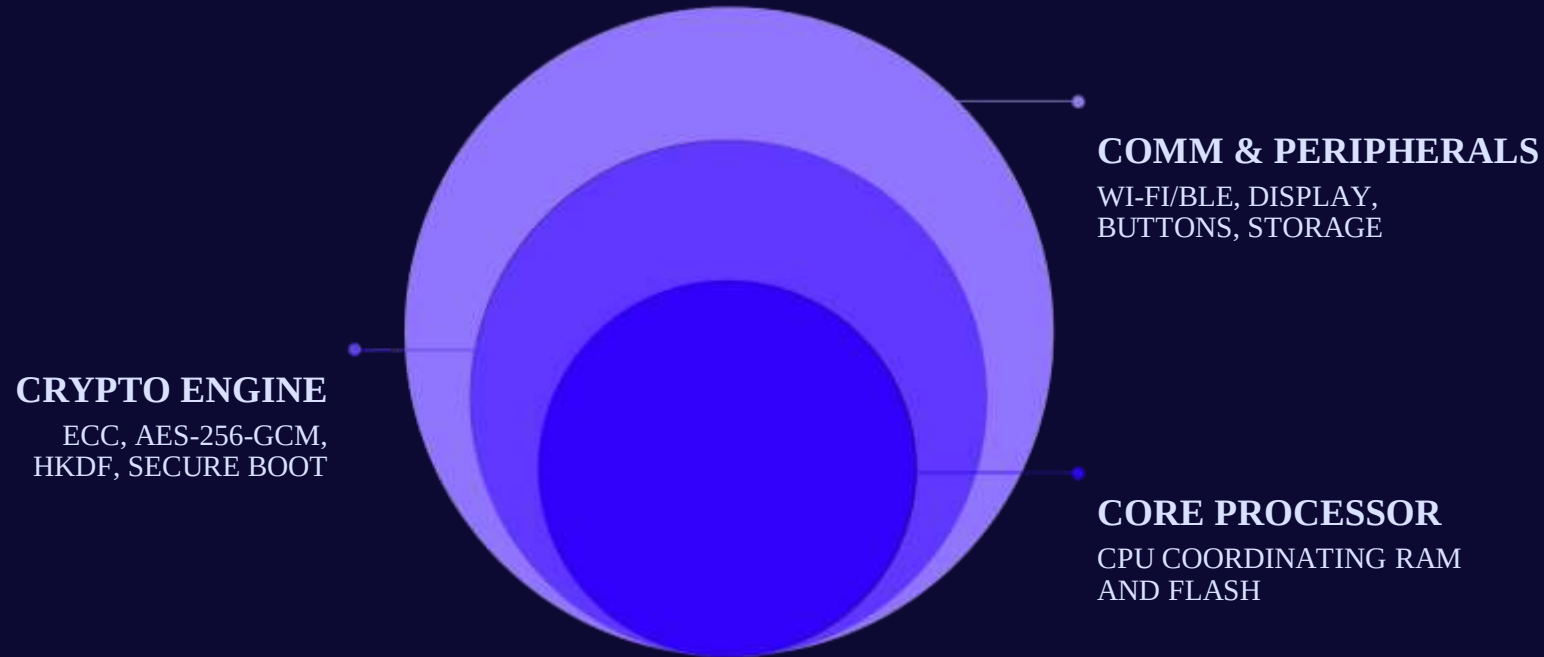


BATTERY-FRIENDLY

MINIMAL CPU CYCLES — PURPOSE-BUILT FOR POWER-CONSTRAINED IOT DEVICES

ARCHITECTURE

SYSTEM ARCHITECTURE & BLOCK DIAGRAM



EACH HARDWARE NODE RUNS A LAYERED CRYPTOGRAPHIC STACK: THE **PROCESSOR** COORDINATES ALL OPERATIONS; THE **CRYPTOGRAPHIC ENGINE** HANDLES ECC, AES-GCM, AND SECURE BOOT; THE **COMMUNICATION LAYER** MANAGES WI-FI/BLE TRANSMISSION; AND **PERIPHERALS** PROVIDE REAL-TIME USER FEEDBACK AND PERSISTENT SECURE STORAGE.

❏ **DATA FLOW:** INITIALIZATION → KEY GENERATION → ECDH KEY EXCHANGE → HKDF SESSION KEY DERIVATION → AES-256-GCM ENCRYPTION → TRANSMISSION → DECRYPTION & VERIFICATION → STATUS DISPLAY

REQUIREMENTS

HARDWARE & SOFTWARE REQUIREMENTS

HARDWARE COMPONENTS

Component	Specification	Purpose
Raspberry Pi 4B	Quad-core ARM @ 1.5 GHz, 4 GB RAM, 32 GB MicroSD	Primary processing unit; sufficient for full ECC operations
ESP32	Dual-core Xtensa @ 240 MHz, 520 KB SRAM, Wi-Fi/BLE	Ultra-low power; suitable for battery-powered IoT nodes
OLED (SSD1306)	0.96" I²C, 128×64 px, Monochrome	Real-time status display with low power draw
LED Indicators	Red / Green / Blue / Yellow (3–5 mm)	Visual feedback: Red=Fail, Green=Secure, Blue=keygen, Yellow=transfer
EEPROM AT24C256	32 KB, I²C interface	Secure storage of device identity and cryptographic metadata

SOFTWARE STACK

Category	Tool / Library	Purpose
OS / Framework	Raspberry Pi OS 64-bit / ESP-IDF v5.0+	Operating system and development framework
Cryptography	OpenSSL / mbedTLS / PyCryptodome	ECC, ECDH, ECDSA, AES-GCM implementation
Language	Python 3.9+ / C	Application development and embedded firmware
Debugging	Putty / Wireshark / OpenSSH	Serial terminal, packet analysis, remote access



OPERATIONAL FLOW & ALGORITHM

COMPLETE SYSTEM WORKFLOW

1 POWER-ON & SECURE BOOT

FIRMWARE INTEGRITY VERIFIED; ECC DOMAIN PARAMETERS LOADED

2 KEY LOADING & PAIRING

PRIVATE KEY LOADED; USER PRESSES "PAIR"; NODES EXCHANGE PUBLIC KEYS Q_A, Q_B

3 ECDH + HKDF

$S = D_A \cdot Q_B = D_B \cdot Q_A$; SESSION KEY $K = \text{HKDF-SHA256}(S)$

4 ECDSA AUTHENTICATION

CHALLENGE-RESPONSE CONFIRMS MUTUAL IDENTITY BEFORE DATA EXCHANGE

5 ENCRYPTED EXCHANGE

SENDER: $(C, \text{TAG}) = \text{AES-256-GCM}(K, \text{NONCE}, M)$; RECEIVER DECRYPTS & VERIFIES TAG

6 SECURE SHUTDOWN

KEYS D_A, D_B, S, K SECURELY OVERWRITTEN FROM MEMORY; STORAGE SYNCED

ECC ALGORITHM — CORE STEPS

INITIALIZE CURVE

LOAD PARAMETERS (P, A, B, G, N) ON BOTH NODES

GENERATE KEYPAIRS

NODE A: $D_A \in [1, N-1], Q_A = D_A \cdot G$
NODE B: $D_B \in [1, N-1], Q_B = D_B \cdot G$

DERIVE SHARED SECRET

$S_A = D_A \cdot Q_B = S_B = D_B \cdot Q_A = D_A \cdot D_B \cdot G$

ENCRYPT & TRANSMIT

TRANSMIT $(\text{NONCE}, C, \text{TAG})$;
PEER DECRYPTS AND
VALIDATES

CLEAR MEMORY

SECURELY OVERWRITE ALL SENSITIVE VARIABLES

❏ **COMPLEXITY:** TIME $O(\log N) \cdot \text{SPACE } O(1)$ WITH <200 BYTES CRYPTOGRAPHIC STATE

IMPLEMENTATION PLAN

Project Overview

14-WEEK IMPLEMENTATION TIMELINE

TIMELINE	PHASE	KEY DELIVERABLE
WEEK 1–2	PLANNING & INITIAL REVIEW	PLATFORM DECISION FINALIZED; INITIAL REVIEW SUBMITTED
WEEK 3–4	ENVIRONMENT & FIRMWARE SETUP	WORKING SKELETON BUILD ON BOTH NODES
WEEK 5–6	ECC KEY GENERATION	VERIFIED ON-DEVICE KEY-PAIR GENERATION
WEEK 7–8	SECURE KEY EXCHANGE	REVIEW 2: MATCHING SHARED SECRET CONFIRMED ACROSS BOTH NODES
WEEK 9–10	AES-GCM ENCRYPTION	END-TO-END ENCRYPTED MESSAGE EXCHANGE DEMONSTRATED
WEEK 11	SECURE BOOT & ERROR HANDLING	FAULT-TOLERANT BOOT WITH COMPREHENSIVE ERROR REPORTING
WEEK 12	SYSTEM INTEGRATION	FULLY INTEGRATED DUAL-NODE PROTOTYPE OPERATIONAL
WEEK 13	TESTING & BENCHMARKING	BENCHMARK REPORT; SECURITY TEST CASES VALIDATED
WEEK 14	FINAL REVIEW	FINAL REVIEW: LIVE DEMO + COMPLETE PROJECT REPORT



INITIAL REVIEW — WEEK 2

BLOCK DIAGRAM, HW/SW REQUIREMENTS, TIMELINE SUBMITTED



REVIEW 2 — WEEK 8

FUNCTIONAL ECDH WITH VERIFIED SHARED SECRET ACROSS BOTH NODES



FINAL REVIEW — WEEK 14

FULLY INTEGRATED, BENCHMARKED, AND DEMONSTRATED PROTOTYPE



ADVANTAGES, APPLICATIONS & FUTURE ENHANCEMENTS

KEY ADVANTAGES

-  **100–1000× FASTER**
100–1000× FASTER THAN RSA
-  **90% SMALLER KEYS**
256-bit ECC = 3072-bit RSA — Nist - Approved P-256 Curves
-  **BATTERY-OPTIMIZED**
MINIMAL CPU CYCLES AND MEMORY FOOTPRINT FOR IOT DEVICES
-  **REPLAY ATTACK RESISTANT**
NONCE-BASED ENCRYPTION AND TIMESTAMP VALIDATION
-  **DUAL-PLATFORM**
RUNS ON BOTH RESOURCE-CONSTRAINED ESP32 AND FULL RASPBERRY PI

REAL-WORLD APPLICATIONS

FUTURE ENHANCEMENTS

- **POST-QUANTUM CRYPTOGRAPHY:** CRYSTALS-DILITHIUM, SPHINCS+ LATTICE-BASED SCHEMES
- **HARDWARE SECURITY MODULE:** TPM 2.0 / ARM TRUSTZONE INTEGRATION
- **MULTI-DEVICE MESH NETWORKS:** DISTRIBUTED KEY MANAGEMENT
- **ML ANOMALY DETECTION:** REAL-TIME BREACH DETECTION
- **FIPS 140-3 CERTIFICATION:** INDUSTRY STANDARDIZATION AND COMPLIANCE



SMART HOMES & IOT

SECURE INTER-DEVICE



MEDICAL DEVICES

ECC $O(\log N)$ VS. RSA $O(K^3)$ FOR EQUIVALENT SECURITY LEVEL



INDUSTRIAL AUTOMATION



CONNECTED VEHICLES

V2V AND V2I SECURE COMMUNICATION



CONCLUSION

CONCLUSION & EXPECTED OUTCOMES

THIS PROJECT DEMONSTRATES A LIGHTWEIGHT, CRYPTOGRAPHICALLY SECURE AUTHENTICATION AND ENCRYPTION SYSTEM TAILORED FOR RESOURCE-CONSTRAINED EMBEDDED DEVICES. BY LEVERAGING ECC WITH 256-BIT KEYS, THE SYSTEM ACHIEVES **SECURITY EQUIVALENT TO 3072-BIT RSA** WHILE CONSUMING A FRACTION OF COMPUTATIONAL AND MEMORY RESOURCES.

 COMPLETE ECDH PROTOCOL FULL ECC-BASED KEY EXCHANGE DESIGNED AND IMPLEMENTED	 AES-256-GCM ENCRYPTION AUTHENTICATED ENCRYPTION WITH INTEGRITY VERIFICATION INTEGRATED
 ECDSA AUTHENTICATION CHALLENGE-RESPONSE SIGNATURES PREVENT SPOOFING AND REPLAY	 SECURE BOOT FIRMWARE INTEGRITY VERIFIED CRYPTOGRAPHICALLY BEFORE EVERY EXECUTION
 DUAL-PLATFORM DEPLOY VALIDATED ON BOTH ESP32 AND RASPBERRY PI HARDWARE	 O(LOG N) COMPLEXITY UNDER 200 BYTES CRYPTOGRAPHIC STATE — TRULY EMBEDDED-READY

AS IOT AND EMBEDDED SYSTEMS PROLIFERATE ACROSS CRITICAL INFRASTRUCTURE, HEALTHCARE, AND CONSUMER APPLICATIONS, THIS PROJECT BRIDGES THE GAP BETWEEN THEORETICAL CRYPTOGRAPHY AND PRACTICAL EMBEDDED CONSTRAINTS — PROVIDING A REFERENCE IMPLEMENTATION ADAPTABLE FROM SMART HOMES TO INDUSTRIAL AUTOMATION AND MEDICAL DEVICES.

THANK YOU — VELLORE INSTITUTE OF TECHNOLOGY, CHENNAI CAMPUS · BCSE305L EMBEDDED SYSTEMS ·

